Journal of Labor Research (2026) 47:4
https://doi.org/10.1007/s12122-026-09381-y

Occupational Preferences Heterogeneity in Labor Supply: Evidence from the PSID

Seyedmasood Dastan[1] · **Faraz Farhidi**[2]

Received: 10 May 2025 / Accepted: 8 March 2026 / Published online: 19 March 2026
© The Author(s) 2026

Abstract
Empirical research generally finds that wage fluctuations lead to only modest adjustments in hours worked. However, individuals experiencing economic hardship often increase their labor supply when wages decline, exhibiting negative labor supply elasticity, a phenomenon documented in both less industrialized economies and certain high-income countries. Using micro-level data from the Panel Study of Income Dynamics (PSID) spanning 1968 to 2017, this study investigates heterogeneity in labor supply elasticity across occupations and industries, with particular attention to the influence of work-environment characteristics and non-pecuniary job attributes. Our findings reveal substantial variation in Frisch elasticity that is systematically associated with occupational characteristics and individual work preferences. Workers in construction, extraction, production, and transportation occupations display relatively high labor supply elasticity, indicating greater responsiveness to wage fluctuations and a stronger propensity to substitute leisure for labor. In contrast, individuals employed in the arts, sports, and social services exhibit lower elasticity estimates, consistent with the hypothesis that these occupations provide higher intrinsic satisfaction or other non-pecuniary returns, thereby reducing sensitivity to wage changes. We further observe cross-state variation in Frisch elasticities that reflect underlying institutional differences in labor market regulation, unionization, and employment protection. Overall, the evidence highlights the importance of occupational heterogeneity and non-pecuniary job characteristics in shaping labor supply behavior at the intensive margin. These results contribute to a deeper understanding of the mechanisms underlying labor supply responsiveness and underscore the need to account for non-monetary job attributes when evaluating labor market dynamics.

Keywords Intertemporal labor substitution · Industrial and occupational heterogeneity · Work preferences

JEL Classifications C15 · C23 · J22 · J32

Extended author information available on the last page of the article

Introduction

A significant advancement in the study of labor supply over the past decades has been the refinement of theoretical frameworks and empirical methodologies to better capture the diverse motivations underlying individuals' work-hour decisions. The lifecycle labor supply model provides a foundational perspective, delineating four key dimensions of labor supply choices: (Altman 2014) the average hours worked over the lifecycle (Altonji 1986), the age-related trajectory of working hours (Andolfatto and Gomme 1996), aggregate fluctuations in labor supply, and (Autor et al. 2013) individual-specific deviations from lifecycle patterns. These components are intrinsically linked through intertemporal substitution effects and wealth effects (Card, 1991). While intertemporal labor substitution is a fundamental element of macroeconomic analysis, empirical evidence remains inconclusive regarding individuals' willingness to adjust labor supply in response to wage variations over time. Most studies indicate that wage fluctuations result in only minor, positive changes in hours worked. Conversely, individuals in economically precarious conditions tend to increase their labor hours in response to declining wages to stabilize their income, exhibiting negative labor supply elasticity.

Altman (2014) argues that, unlike commodities, human labor supply is not merely a function of wages but is shaped by sociological, psychological, and institutional factors that influence effort levels and working hours. He identifies three principal determinants in this decision-making process: the self, personal aspirations, and the external environment. Kaufman (1999) further asserts that individuals pursue labor supply choices that align with their perceived optimal outcomes, constrained by environmental conditions and the inherent uncertainty of economic variables.

The Frisch elasticity of labor supply plays a crucial role in shaping macroeconomic policies, particularly in the evaluation of Social Security reforms (Imrohoroglu & Kitao, 2012) and the determination of optimal capital and labor tax rates (Peterman, 2013). Furthermore, labor supply elasticity is instrumental in assessing the broader implications of fiscal policy, taxation, and government spending (Auerbach & Kotlikoff, 1987; Judd, 1987). Jaimovich and Siu (2009) highlight the impact of demographic shifts on labor supply volatility, suggesting that these factors account for nearly one-third of the Great Moderation. Several studies incorporate heterogeneity—accounting for variations in family composition, labor market fluctuations and wealth effect—into aggregate labor supply models that examine intertemporal labor-leisure trade-offs (Kydland, 1984; Cho & Rogerson, 1988; Andolfatto & Gomme, 1996; Castaneda et al., 1998; Merz, 1999; den Haan et al., 2000; Gomes et al., 2001; Chang & Kim, 2006, 2007). These works primarily focus on the cyclical behavior of unemployment rates as well as heterogeneity in household wealth and labor supply (Mustre-del-Rio, 2015).

A substantial body of research suggests that estimates of intertemporal labor supply elasticity may be biased due to liquidity constraints (Zeldes, 1989; Ziliak & Kniesner, 1999; Pistaferri, 2003; French, 2005; Domeij & Floden, 2006; Guvenen, 2006; Contreras & Sinclair, 2008; Chang et al., 2011; Chetty, 2012; Rogerson & Wallenius, 2013). Imai and Keane (2004) emphasize that human capital investment is age-dependent, necessitating its inclusion in elasticity estimations. Furthermore, studies

by Chang and Kim (2006), Gourio and Noual (2009), Fiorito and Zanella (2012), and Peterman (2016) attempt to reconcile discrepancies between microeconomic and macroeconomic elasticity estimates, underscoring the role of gender, marital status, secondary earnings, and age in shaping labor supply decisions.

Whalen and Reichling (2017) provide additional insights into the determinants of labor supply elasticity. Their findings suggest that: (Altman 2014) men exhibit lower Frisch elasticity than women, implying that men adjust their work hours less in response to changes in current or future wages; (Altonji 1986) younger individuals have lower elasticity than those approaching retirement; (Andolfatto and Gomme 1996) women without children exhibit lower elasticity compared to women with children; and (Autor et al. 2013) dual-income households are more responsive to wage fluctuations than single-earner households.

Given the variability in prior elasticity estimates, we posit that these findings may be systematically biased due to interactions between occupational selection and wealth constraints, which in turn shape individuals' labor supply preferences. Consequently, our study aims to empirically examine deviations in Frisch elasticity by incorporating occupational heterogeneity and wealth effects into the analysis.

To achieve this objective, we estimate key labor supply parameters using PSID data from 1968 to 2017, stratified by wealth levels, age cohorts, and occupational classifications. Our methodology is based on a customized log-linearization of the household Euler equation, adapted from Domeij and Floden (2006). The findings indicate that workers in healthcare and public administration exhibit lower responsiveness to wage fluctuations compared with those in agriculture, fishing, and manufacturing. This reduced elasticity appears to reflect the greater non-monetary fulfillment, job satisfaction, and work flexibility associated with occupations in these sectors.

This study contributes to the literature by offering empirical evidence on the role of occupational preferences and wealth constraints in shaping labor supply elasticity. The results carry significant policy implications, particularly for taxation frameworks, workforce regulations, and employment benefits.

Model (Fig. 1)

The figure above illustrates the core hypothesis we advance in this study: variations in working hours are shaped not only by differences in wages but also by heterogeneity in work environments. In particular, we posit that labor supply at the intensive

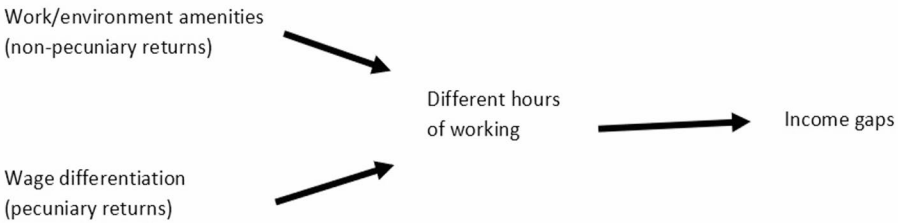

Fig. 1 Conceptual model which hypothesizes the main argument

margin reflects both pecuniary and non-pecuniary considerations. Accordingly, an individual may increase or decrease the number of hours they choose to work not solely in response to changes in wages, but also based on the non-monetary benefits associated with their occupation—such as job satisfaction, enjoyment, autonomy, or a sense of purpose and fulfillment.

Under this framework, observed income disparities need not arise exclusively from wage differentials. Instead, they may partly reflect systematic variation in the quality of work environments across occupations and industries. If certain occupations offer substantially higher non-pecuniary returns, individuals in these roles may choose to supply more hours or maintain stable hours despite equivalent wages, resulting in lower responsiveness to wage fluctuations. Conversely, individuals in less fulfilling occupations may reduce their hours or adjust them more readily in response to wage changes, reflecting higher sensitivity to pecuniary incentives and compensation for lower intrinsic utility. This mechanism yields a clear testable implication: if non-pecuniary job attributes meaningfully influence labor supply decisions, the estimated Frisch elasticity should vary systematically across occupations and industries in ways that align with differences in job satisfaction and other measures of work-related fulfillment.

Under this hypothesis, heterogeneity in work environments—rather than wage dispersion alone—may serve as one of the fundamental drivers of income and wealth inequality within an economy. To the best of our knowledge, this channel has received limited attention in existing literature, and our study contributes to filling this gap by providing both theoretical motivation and empirical evidence supporting the relevance of non-pecuniary factors in explaining observed labor supply behavior.

The proposed model incorporates a joint household (HH) utility function, where L represents leisure and C denotes consumption. In this framework, we introduce heterogeneous labor preferences, recognizing that individuals exhibit varying leisure preferences based on their working environment. The model utilizes Frisch elasticity to capture these differences while maintaining standard utility function properties, including well-behaved and quasi-concave characteristics. The household optimizes utility subject to time constraints and wealth distribution.

A subsistence or survival consumption constraint is imposed, requiring that households earn a minimum income necessary to cover fundamental needs such as food and shelter. In contrast, the budget constraint dictates that, in the absence of savings, total consumption must equal total income. Here, W*h represents market income, while “a” denotes nonmarket income (e.g., asset returns). For simplicity, we assume that the household’s total nonmarket assets remain constant and are transferred across generations. In other words, the net inflow and outflow of assets sum to zero over time.

The subsistence wage (SW) is determined by the interaction between time constraints and subsistence consumption requirements. At this threshold, household members must allocate all available working hours to meet their basic needs. However, due to data limitations, an empirical investigation of this phenomenon remains challenging. Below the subsistence income level, individuals must prioritize survival, compelling them to work under precarious conditions with limited choice, thereby eliminating the possibility of substituting leisure for labor. As a result, the indifference curve transitions from a U-shape to an L-shape, reflecting a rigid labor supply behavior.

In extreme cases, wage earners who fall below the subsistence threshold may be unable to sustain the physical demands of a full workday, potentially forcing their families into alternative survival strategies such as begging, informal labor, or illicit activities. This dynamic leads to "forced unemployment" or engagement in crime as a means of economic survival. Conversely, when monetary income significantly exceeds subsistence levels, the minimum income constraint ceases to be binding, allowing individuals greater flexibility in their labor-leisure choices. Under such circumstances, households optimize consumption and labor supply to maximize their expected discounted utility:

$$U_0 = E_0 \sum_{t=0}^{\infty} \beta^t u(c_t, h_{it})$$

Where β is the subjective discount factor.

$$u\left(c_t, h_{it}\right) = \frac{c_t^{1-\mu}}{1-\mu} - \alpha \frac{h_{it}^{1+1/\gamma_i}}{1+1/\gamma_i}$$

γ_i is the Frisch elasticity for different job occupations (or industries). The individuals' budget constraint in period t is given by:

$$\mathrm{c_t} + \mathrm{a_{t+1}} \leq (1 + \mathrm{r})\, \mathrm{a_t} + \mathrm{w_t h_{it}}$$

'r' is the return on savings. The distribution of households in wealth–wage space is constant; however, there might be fluctuations and uncertainty at the household level. We also assume that the logged wage derived from an AR (Altman 2014) process with fixed effects

$$lnw_{it} = \psi_i + z_{it} and z_{it} = \rho z_{it-1} + \epsilon_{it}$$

.

The subsistence constraint would be as follows:

$$SW_{it} = a_t + w_t h_{it} + \bar{a_t}$$

a_t is the amount that they can borrow in each period. If an individual at the margin of the subsistence level, faces a negative and exogenous wage shock if she can borrow from any sources, she might not substitute her leisure with hours of working; otherwise, she must.

Various authors have estimated similar stochastic processes for logged labor productivity using data from the PSID which we are going to take advantage of their results, such as work by Card (1991), Hubbard et al. (1994), Flodén and Lindé (2001), and Domeij & Floden (2006). Solving for the first-order conditions, we get to

different equations to estimate (used by Domeij & Floden (2006), and by McCurdy (1981) and Altonji (1986):

$$\Delta \ln(h_{t+1}) = constant + \gamma_i \Delta \ln(w_{t+1}) - \frac{\gamma_i \Phi_t}{\lambda_t} + \gamma_i \epsilon_{t+1} \tag{1}$$

$$\ln(h_t) = constant + \gamma_i \ln(w_t) - \frac{\gamma_i}{\mu} \ln(c_t) \tag{2}$$

The base equation we use is a log-linearization of the household's Euler equation:

$$\Delta \ln(h_{t+1}) = constant + \gamma_i \Delta \ln(w_{t+1}) + \epsilon_{t+1} \tag{3}$$

We adopt the above equation, by controlling for the key observables—provided in the dataset—and then adapting the IV approach using the two wage lags to predict the wage—suggested by the literature—to measure the Frisch elasticity as follows:

$$\Delta \ln(w_{ijt}) = \alpha_{1j} + \delta_{1t} + \beta_{1i} \Delta \ln(w_{ijt-1}) + \beta_{2i} \Delta \ln(w_{ijt-2}) + \theta_k X_{ijkt} + \epsilon_{1ijt} \tag{4}$$

$$\Delta \ln(h_{ijt}) = \alpha_{2j} + \delta_{2t} + \gamma_i \Delta \widehat{\ln(w_{ijt})} + \phi_k X_{ijkt} + \epsilon_{2ijt} \tag{5}$$

Based on Eqs. 4 and 5, we can test the hypothesis, which is the impact of the changes in occupation (i) on the Frisch elasticities (γ_i) at different wealth and age levels. In the above equations, $\Delta \ln(h_{ijt})$ is the difference in log of working hours for individual j, at time t, and work in an occupation (industry) i. $\Delta \ln(w_{ijt})$ is the difference in log wages. α_j is the individual's fixed effect, δ_t is the time fixed effect, X_{ijkt} are the controls—age, age-square, education, education-square, wife's income, and other household sources' income.

Data

This study utilizes data from the Panel Study of Income Dynamics (PSID), covering the period from 1968 to 2017. The dataset provides a comprehensive longitudinal perspective on labor supply behavior and household economic conditions. The summary statistics are presented in the table below. In addition to the primary variables of interest—including various wage measures and the working hours of household heads—we incorporate key covariates such as age and educational attainment to account for demographic and human capital factors. Furthermore, we include food costs as a proxy for household living expenses, as this variable serves as an indicator of individuals' expectations regarding their standard of living, which may, in turn, influence labor supply decisions.

It is important to note that this study focuses on variations in hours worked rather than movements into or out of employment. Accordingly, the primary locus of analysis is the intensive margin of labor supply. To ensure that the sample reflects indi-

Table 1 Summary statistics of the main variables of the study

Variables	Number of Observations	Mean	Median	Std. Dev.	Min	Max
Head wage	74,075	36,316	25,000	58,645	15	3,500,000
Head Salary	74,075	35,610	10,000	79,268	10	3,200,000
Head other income	74,075	3,656	45	27,621	0	5,000,000
Wife income	74,075	10,081	0	42,190	0	9,999,996
Head age	74,075	39.45	38	11.75	16	99
Head educational level	73,114	9.32	7	5.198	0	17
Food cost	74,075	1,882	1,560	1,986	0	67,600
Food stamp value	74,075	49.02	0	494.13	0	9,999
Head average hours of working	73,997	31.61	40	21.964	0	112

Source: PSID

viduals who are continuously engaged in paid employment, all observations in which the household head reports zero wage or salary income have been excluded. This restriction allows the analysis to concentrate on adjustments in labor intensity among active workers, thereby minimizing confounding effects arising from employment transitions or temporary nonparticipation (Table 1).

Result and discussion

It is important to emphasize at the outset that the Frisch elasticity, as estimated in our framework, captures the responsiveness of an individual's hours of work to changes in wages while holding marginal utility of wealth constant. Individuals with higher Frisch elasticities are, in this context, comparatively more willing to substitute leisure for labor when wages change. However, within the theoretical framework introduced earlier, a higher estimated Frisch elasticity may also indicate that these individuals derive lower non-pecuniary satisfaction from their work environment. Consequently, they require a larger wage-induced incentive to adjust their hours. In contrast, individuals with lower Frisch elasticities may obtain greater intrinsic satisfaction, fulfillment, or utility from their work, and thus exhibit weaker responsiveness to wage fluctuations at equivalent wage and hour levels.

It is also essential to reiterate that the focus of our empirical analysis is the intensive margin of labor supply—that is, adjustments in the number of hours worked among individuals who are already participating in the labor market. Our estimates therefore reflect variation in labor intensity rather than the binary decision to enter or exit employment. This distinction is central to interpreting the results that follow, as the intensive margin is where non-pecuniary work characteristics are most plausibly manifested in observable labor supply behavior.

Table 2 presents Frisch elasticity estimates for the full sample, as well as for subsamples categorized by household savings levels. The findings indicate that individuals with higher savings levels tend to derive greater enjoyment from work compared to those with little or no savings.

Table 2 The main analysis based on different saving rates

	(1)	(2)	(3)	(4)
	Full sample	Saving > 0	Saving > 1,000	Saving > 10,000
Frisch elasticity	0.280***	0.228***	0.218***	0.210***
	(0.009)	(0.016)	(0.019)	(0.034)
Controls*	Yes	Yes	Yes	Yes
Fixed Effects**	Yes	Yes	Yes	Yes
N	42,938	19,242	14,054	5,799
R^2	0.618	0.487	0.516	0.552

Standard errors in parentheses

$^{*}p<0.05$, $^{**}p<0.01$, $^{***}p<0.001$

*Controls: Age of the head, square age, head education, square education, wife income, head other income, households' food cost, and households' food stamp received. Duration: 1968–2017

** Individual, time, industrial and occupational fixed effects

Table 3 The main analysis based on different saving rates: Using Salary as the dependent variable

	(1)	(2)	(3)
	Full sample	Saving > 0	Saving > 1,000
Frisch elasticity	-0.011	-0.045	0.022
	(0.068)	(0.106)	(0.153)
Controls*	Yes	Yes	Yes
Fixed Effects**	Yes	Yes	Yes
N	1,163	851	729
R^2	0.482	0.453	0.362

Standard errors in parentheses

$^{*}p<0.05$, $^{**}p<0.01$, $^{***}p<0.001$

*Controls: Age of the head, square age, head education, square education, wife income, head other income, households' food cost, and households' food stamp received. Duration: 1968–2017

** Individual, time, industrial and occupational fixed effects

The results for salaried households, presented in Table 3, show that the estimated Frisch elasticities are statistically insignificant and fluctuate around zero. This stands in clear contrast to the patterns observed among hourly wage earners, for whom the elasticity estimates are both economically meaningful and statistically distinguishable from zero. The near-zero estimates in the salaried subsample suggest that hours worked by these individuals are largely unresponsive to environment variation—consistent with the view that salaried positions may embody stronger contractual rigidity or higher levels of non-pecuniary compensation. Taken together, these findings provide cautious yet meaningful support for the validity of our central hypothesis, namely that heterogeneity in work environments and non-pecuniary returns play an important role in shaping labor supply behavior at the intensive margin.

Tables 4 and 5 report elasticity estimates across wage and age groups within savings categories. While the point estimates for lower-wage households are generally smaller than those for higher-wage households, the associated standard errors are sufficiently large that differences are not always statistically distinguishable across groups. Formal Wald tests of equality between low- and high-wage elasticities fail to

Table 4 The main analysis based on different age and wealth groups: households with some savings

	(1)	(2)	(3)	(4)	(5)
	Saving > 0	Saving > 0 W < 25,000	Saving > 0 W > = 25,000	Saving > 0 Age < 40	Saving > 0 Age > = 40
Frisch elasticity	0.228***	0.190***	0.224***	0.232***	0.238***
	(0.016)	(0.049)	(0.020)	(0.031)	(0.027)
Controls*	Yes	Yes	Yes	Yes	Yes
Fixed Effects**	Yes	Yes	Yes	Yes	Yes
N	19,242	5,864	12,972	9,460	7,766
R^2	0.487	0.371	0.555	0.452	0.541

Standard errors in parentheses

$^{*}p<0.05$, $^{**}p<0.01$, $^{***}p<0.001$

*Controls: Age of the head, square age, head education, square education, wife income, head other income, households' food cost, and households' food stamp received. Duration: 1968–2017

** Individual, time, industrial and occupational fixed effects

Table 5 The main analysis based on different age and wealth groups: households with more than 1,000 $ savings

	(1)	(2)	(3)	(4)	(5)
	Saving > 1000	Saving > 1000 W < 25,000	Saving > 1000 W > 25,000	Saving > 1000 Age < 40	Saving > 1000 Age > 40
Frisch elasticity	0.218***	0.101	0.222***	0.239***	0.238***
	(0.019)	(0.081)	(0.022)	(0.039)	(0.027)
Controls*	Yes	Yes	Yes	Yes	Yes
Fixed Effects**	Yes	Yes	Yes	Yes	Yes
N	14,054	3,178	10,876	6,288	7,766
R^2	0.516	0.376	0.559	0.484	0.541

Standard errors in parentheses

$^{*}p<0.05$, $^{**}p<0.01$, $^{***}p<0.001$

*Controls: Age of the head, square age, head education, square education, wife income, head other income, households' food cost, and households' food stamp received. Duration: 1968–2017

** Individual, time, industrial and occupational fixed effects

reject the null of equal coefficients in several specifications. Accordingly, the evidence should be interpreted as suggestive rather than definitive. The directional pattern is consistent with the hypothesis that lower-income individuals face tighter constraints or limited marginal returns to additional hours, but the magnitude differences should be viewed cautiously.

When segmenting the sample by age groups—those below and above forty—we observe a less consistent pattern compared to wealth effect. The results suggest that younger workers tend to enjoy their jobs barely more than older individuals, or alternatively, they may be relatively more ambitious in their career aspirations.

Table 6 presents Frisch elasticity estimates across major industry sectors, offering insight into how occupational environments shape labor supply responsiveness. The results indicate that individuals employed in healthcare and public administration exhibit lower Frisch elasticity, suggesting that these positions may provide greater non-pecuniary returns—such as intrinsic job satisfaction, flexibility, autonomy, or a sense of purpose—that reduce workers' willingness to adjust their

Table 6 The main analysis based on different industries

	(1)	(2)	(3)	(4)	(5)	(6)
	Full sample	Agriculture, Fishing, Forestry	Utilities	Healthcare	Manufacturing	Public Administration and Active Military
Frisch elasticity	0.280***	0.579***	0.384***	0.131*	0.302***	0.212**
	(0.009)	(0.156)	(0.059)	(0.066)	(0.024)	(0.075)
Controls*	Yes	Yes	Yes	Yes	Yes	Yes
Fixed Effects**	Yes	Yes	Yes	Yes	Yes	Yes
N	42,938	1,489	3,428	3,460	11,392	3,268
R^2	0.618	0.466	0.602	0.601	0.662	0.531

Standard errors in parentheses

$^{*}p<0.05$, $^{**}p<0.01$, $^{***}p<0.001$

*Controls: Age of the head, square age, head education, square education, wife income, head other income, households' food cost, and households' food stamp received. Duration: 1968–2017

** Individual, time, industrial (for the first regression), and occupational fixed effects

hours in response to wage fluctuations. By contrast, workers in agriculture, fishing, and forestry display substantially higher elasticity estimates, implying a greater readiness to substitute leisure for labor. This pattern is consistent with expectations, as these occupations are often characterized by demanding physical labor, elevated occupational risk, rigid scheduling, and more challenging work environments. Such attributes may lower job satisfaction and make workers more responsive to wage changes, increasing the likelihood that they supply additional hours only when sufficiently compensated.

Table 7 further disaggregates the analysis by examining Frisch elasticity across detailed occupational categories, thereby allowing us to investigate within-industry variation in labor supply responsiveness. While the results indicate meaningful differences in elasticity estimates across occupations, these findings must be interpreted with caution. Several occupational groups contain relatively small samples, which reduces statistical power and may introduce estimation imprecision or bias. Additionally, heterogeneity in occupational characteristics—such as job security, autonomy, physical demands, or sector-specific labor market constraints—may independently influence workers' sensitivity to wage fluctuations, potentially contributing to the observed dispersion in elasticity values.

Despite these limitations, the overall pattern remains consistent with our theoretical framework. Occupations associated with higher intrinsic enjoyment or fulfillment—such as artistry, sports, and sales—tend to exhibit lower Frisch elasticity values, suggesting weaker responsiveness to wage changes and perhaps greater non-pecuniary returns. In contrast, occupations characterized by physically demanding or hazardous working conditions, including construction, extraction, transportation, and production, display relatively higher elasticity estimates. This divergence aligns with the hypothesis that non-pecuniary job attributes systematically shape labor supply behavior at the intensive margin.

Table 7 The main analysis based on different occupations

	(1)	(2)	(3)	(4)	(5)	(6)
	Full sample	Business Operations and Management	Food, Cleaning, Maintenance Services	Sales Occupations	Art, Education, Sport, and Social Services	Construction, Extraction, Production,and Transportation
Frisch elasticity	0.280***	0.287***	0.249***	0.219*	0.148	0.338***
	(0.009)	(0.056)	(0.032)	(0.092)	(0.083)	(0.016)
Controls*	Yes	Yes	Yes	Yes	Yes	Yes
Fixed Effects**	Yes	Yes	Yes	Yes	Yes	Yes
N	42,938	4,033	7,791	2,566	1,751	19,869
R^2	0.618	0.621	0.608	0.539	0.522	0.628

Standard errors in parentheses

$^{*}p<0.05$, $^{**}p<0.01$, $^{***}p<0.001$

*Controls: Age of the head, square age, head education, square education, wife income, head other income, households' food cost, and households' food stamp received. Duration: 1968–2017

** Individual, time, industrial, and occupational (for the first regression) fixed effects

Table 8 The main analysis, based unionization

	(1) Full sample	(2) If a job covers by a Union	(3) If a job does not cover by a Union
Frisch elasticity	0.280***	0.284***	0.279***
	(0.009)	(0.028)	(0.013)
Controls*	Yes	Yes	Yes
Fixed Effects**	Yes	Yes	Yes
N	42,938	8,814	27,860
R^2	0.618	0.680	0.608

Standard errors in parentheses

$^{*}p<0.05$, $^{**}p<0.01$, $^{***}p<0.001$

*Controls: Age of the head, square age, head education, square education, wife income, head other income, households' food cost, and households' food stamp received. Duration: 1968–2017

** Individual, time, industrial and occupational fixed effects

Table 8 presents a comparative analysis of unionized versus non-unionized employment, examining differences in Frisch elasticity and workplace conditions. The findings indicate that union membership does not necessarily correlate with higher job satisfaction or reduced workplace stress. This suggests that while unionized jobs often provide greater job security, collective bargaining power, and better benefits, these advantages may not directly translate into perceived improvements in job satisfaction or lower work-related pressure.

However, when analyzing Frisch elasticity in the context of overtime work, we observe that unionized workers exhibit, on average, a slightly higher (although not significant) elasticity than their non-unionized counterparts. This implies that union members may be more responsive to opportunities for additional work hours, potentially due to union-negotiated regular time pay structures or contractual agreements that incentivize extended work hours under specific conditions.

Beyond unionization, we explore an alternative hypothesis regarding the role of state-level labor protections and regulatory policies in shaping labor supply decisions. Specifically, we posit that non-pecuniary job returns, such as workplace safety standards, healthcare benefits, and job security measures, may influence individuals' willingness to work additional hours. These factors, while not explicitly captured in our dataset, are often embedded within state-level labor laws and employment regulations.

To test this hypothesis, we assume that workers in states with weaker labor protections are more inclined to work longer hours, resulting in lower Frisch elasticity—provided that other economic conditions, including wages, cost of living, marital status, and age, remain constant. Table 9 presents Frisch elasticity estimates across selected states, revealing some differences that align with variations in state-level employment protections. These findings suggest that institutional labor policies play a significant role in shaping labor supply responsiveness, reinforcing the need for future research to incorporate policy-driven labor market heterogeneity into elasticity estimations.

Table 9 Estimating Frisch elasticities for selected states

	(1) California	(2) Tennessee & Kentucky	(3) Massachusetts & New York	(4) Iowa & Nebraska	(5) Texas
Frisch elasticity	0.495***	0.505**	0.435*	0.341**	0.312*
	(0.128)	(0.165)	(0.165)	(0.122)	(0.128)
Controls*	Yes	Yes	Yes	Yes	Yes
Fixed Effects**	Yes	Yes	Yes	Yes	Yes
N	1,844	1,638	1,727	1,927	2,119
R^2	0.602	0.394	0.4266	0.559	0.546

Standard errors in parentheses

*$p<0.05$, **$p<0.01$, ***$p<0.001$

*Controls: Age of the head, square age, head education, square education, interaction between age and education, wife income, head other income, households' food cost, and households' food stamp received. Duration: 1968–2017

** Individual, time, industrial, and occupational fixed effects

The cross-state variation in Frisch elasticities reported in Table 9, suggests that institutional labor market environments may help explain differences in intensive-margin responsiveness. Although our specification controls for individual, occupational, and industry fixed effects, state-level labor institutions vary systematically across the selected states and may influence how workers adjust hours in response to wage fluctuations.

A central institutional distinction concerns right-to-work (RTW) legislation and union coverage. Tennessee, Kentucky, Texas, Iowa, and Nebraska are right-to-work states, whereas California, Massachusetts, and New York are not. RTW laws are associated with lower union density and weaker collective bargaining coverage (Holmes, 1998; Farber et al., 2021). Consistent with this, Bureau of Labor Statistics data document substantially higher union membership rates in California, Massachusetts, and New York relative to the Southern and Midwestern RTW states in our sample (BLS, 2025). Lower union density and weaker bargaining institutions may increase wage dispersion and income volatility, potentially amplifying workers' responsiveness to wage shocks along the intensive margin.

California presents a distinct case. Despite not being a right-to-work state, it exhibits relatively high estimated elasticity. California maintains stronger labor protections, including state-level overtime mandates and a minimum wage historically above the federal floor. At the same time, it features a substantially higher cost of living. Elevated subsistence requirements may increase sensitivity to wage fluctuations, particularly for liquidity-constrained households, consistent with models linking income risk and labor supply responsiveness (Chetty, 2012). Thus, California's higher elasticity may reflect the interaction of wage dispersion, regulatory structure, and cost-of-living pressures.

Massachusetts and New York, which display somewhat lower elasticity estimates, combine strong union presence with relatively generous social insurance systems and a larger public-sector employment base. More extensive income-smoothing mechanisms may reduce the necessity-driven component of labor supply adjustment (Autor et al., 2013). This is consistent with their lower responsiveness relative to California and the Southern RTW states.

In contrast, Iowa, Nebraska, and Texas exhibit lower estimated elasticities. These states are characterized by lower cost-of-living levels and industrial compositions with larger shares of agriculture, energy, and construction. In such environments, hours may be more technologically or contractually rigid, limiting the scope for marginal hour adjustments even when wages fluctuate. Lower subsistence costs may further dampen the marginal impact of wage shocks on hours decisions.

Taken together, the ordering of elasticities across states is consistent with the view that institutional labor market structures—including right-to-work status, union density, wage regulation, and social insurance generosity—interact with occupational characteristics to shape intensive-margin labor supply behavior. While we do not interpret these associations as causal, the results suggest that state-level institutional heterogeneity is an important dimension for understanding cross-sectional variation in Frisch elasticities.

An alternative explanation for changes labor supply could be that individuals work additional hours not in response to occupation/industry incentives, but rather as a means of compensating for income loss due to illness. If individuals experiencing health issues or family medical burdens increase their labor supply to offset these financial setbacks, then Frisch elasticity estimates may be biased when failing to account for this behavior.

To empirically test this hypothesis, we employ the Heckman selection approach, which allows us to control for potential sample selection bias by incorporating an unobserved selection mechanism into our estimation. In the first stage, we predict sick days based on key covariates, including age, education, spouse's income, other household income sources, and food costs. These variables serve as proxies for financial stability, health-related expenditures, and socioeconomic background, which could influence an individual's decision to work additional hours despite illness.

In the second stage, we estimate the main labor supply equation while incorporating the Heckman correction term (Heckit error term), which adjusts for potential biases arising from unobserved health-related factors affecting labor decisions. The results, presented in Table 10, reveal that Frisch elasticity becomes significantly

Table 10 Applying Heckman selection to the main analysis: Sick days might be the motivation for working extra hours

	(1)	(2)
	Overtime	Overtime
Frisch elasticity	30.36**	41.39*
	(10.599)	(16.495)
Heckman error	No	399.8
		(252.4)
Controls*	Yes	Yes
Fixed Effects**	Yes	Yes
N	43,214	26,358
R^2	0.008	0.006

Standard errors in parentheses

$^{*}p<0.05$, $^{**}p<0.01$, $^{***}p<0.001$

*Controls: Age of the head, square age, head education, square education, wife income, head other income, households' food cost, and households' food stamp received. Duration: 1968–2017

** Individual, time, industrial and occupational fixed effects

different and higher after accounting for these unobservable factors. This suggests that previous elasticity estimates could understate the true responsiveness of labor supply, as a portion of observed labor hour adjustments was driven by compensatory work behavior due to health-related financial pressures rather than voluntary wage-driven labor substitution. These findings highlight the importance of considering health shocks and financial constraints in labor supply models, reinforcing the need for further research into the interplay between illness, household income stability, and labor elasticity dynamics.

Conclusion

According to standard labor supply models, individuals allocate their working hours primarily in response to income fluctuations, optimizing their labor-leisure trade-off based on wage incentives. However, our analysis reveals that labor supply decisions are influenced not only by financial considerations but also by non-monetary factors related to the work environment. Specifically, job satisfaction, workplace conditions, and intrinsic occupational preferences play a crucial role in shaping individuals' willingness to supply labor, challenging the assumption that earnings alone drive labor supply behavior.

Our findings demonstrate that individuals' preferences toward working hours are significantly shaped by the nature of their occupation, independent of wage or wealth differentials. Employees who derive higher nonpecuniary returns from their jobs—such as those in the healthcare and public administration industries—exhibit lower Frisch elasticity, indicating a lower willingness to adjust labor supply in response to external factors aside from wages. In contrast, workers in physically demanding and hazardous industries—such as agriculture, fishing, and manufacturing—display higher elasticity, suggesting that the disutility of work in these sectors limits their satisfaction in labor supply decisions only to wage fluctuations.

Also, job attribute preferences are directly observable in the data, and we confirm their influence by demonstrating heterogeneous Frisch elasticities across occupations with similar earnings levels. This suggests that employees in more enjoyable and flexible occupations are more inclined to work additional hours—without monetary incentives—, whereas those in less desirable work environments are more constrained in their ability to adjust labor supply without pecuniary return.

These insights have significant implications for labor market policies, workforce management, and compensation structures. Policymakers and employers should recognize that workplace conditions and job satisfaction factors influence labor supply decisions, and wage adjustments alone may not be sufficient to incentivize labor participation across different sectors. Future research should further explore the interaction between job quality, labor flexibility, and economic incentives, incorporating behavioral and institutional factors into traditional labor supply models to provide a more comprehensive understanding of workforce dynamics.

Author Contribution Conceptualization: F.F.; Methodology: F.F., S.D.; Data collection: F.F.; Formal analysis: F.F., S.D; Writing (original draft preparation): F.F., S.D.; Writing (review and editing): F.F., S.D.. All authors have read and agreed to the published version of the manuscript.

Funding This research received no external funding

Data Availability The data that support the findings of this study are publicly available at https://psidonline.isr.umich.edu/

Declarations

Ethical Approval This study used publicly available, fully anonymized secondary data. Therefore, ethical approval was not required

Informed Consent Not applicable. The study involved secondary analysis of publicly available anonymized data

Conflict of interest The authors declare that they have no conflict of interest

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by-nc-nd/4.0/.

References

Altman M (2014) Insights from behavioral economics on how labor markets work

Altonji JG (1986) Intertemporal substitution in labor supply: evidence from micro data. J Political Econ 94(3, Part 2) S176–S215

Andolfatto D, Gomme P (1996) Unemployment insurance and labor-market activity in Canada. In Carnegie-Rochester conference series on public policy, vol. 44. North-Holland, pp. 47–82

Autor DH, Dorn D, Hanson GH (2013) The China syndrome: Local labor market effects of import competition in the United States. Am Econ Rev 103(6):2121–2168

Auerbach AJ, Kotlikoff LJ (1987) Dynamic fiscal policy. Cambridge University Press

Bureau of Labor Statistics (2025) Union membership rates highest in Hawaii and New York; lowest in North Carolina in 2024. U.S. Department of Labor. https://www.bls.gov/opub/ted/2025/union-membership-rates-highest-in-hawaii-and-new-york-lowest-in-north-carolina-in-2024.htm

Card D (1991) Intertemporal labor supply: an assessment (No. w3602). National Bureau of Economic Research

Castaneda A, Dıaz-Giménez J, Rıos-Rull JV (1998) Exploring the income distribution business cycle dynamics. J Monet Econ 42(1):93–130

Chang Y, Kim SB (2006) From individual to aggregate labor supply: A quantitative analysis based on a heterogeneous agent macroeconomy. Int Econ Rev 47(1):1–27

Chang Y, Kim SB (2007) Heterogeneity and aggregation: Implications for labor-market fluctuations. Am Econ Rev 97(5):1939–1956
Chang Y, Kim SB, Kwon K, Rogerson R (2011) Interpreting labor supply regressions in a model of full- and part-time work. Am Econ Rev 101(3):476–481
Chetty R (2012) Bounds on elasticities with optimization frictions: A synthesis of micro and macro evidence on labor supply. Econometrica 80(3):969–1018
Cho JO, Rogerson R (1988) Family labor supply and aggregate fluctuations. J Monet Econ 21(2–3):233–245
Contreras J, Sinclair S (2008) Labor supply response in macroeconomic models: assessing the empirical validity of the intertemporal labor supply response from a stochastic overlapping generations model with incomplete markets
Den Haan WJ, Ramey G, Watson J (2000) Job destruction and propagation of shocks. Am Econ Rev 90(3):482–498
Domeij D, Floden M (2006) The labor-supply elasticity and borrowing constraints: Why estimates are biased. Rev Econ Dyn 9(2):242–262
Farber HS, Herbst D, Kuziemko I, Naidu S (2021) Unions and inequality over the twentieth century: New evidence from survey data. Q J Econ 136(3):1325–1385
Fiorito R, Zanella G (2012) The anatomy of the aggregate labor supply elasticity. Rev Econ Dyn 15(2):171–187
Flodén M, Lindé J (2001) Idiosyncratic risk in the US and Sweden: Is there a role for government insurance? Rev Econ Dyn 4(2):406–437
French E (2005) The effects of health, wealth, and wages on labour supply and retirement behaviour. Rev Econ Stud 72(2):395–427
Gomes J, Greenwood J, Rebelo S (2001) Equilibrium unemployment. J Monet Econ 48(1):109–152
Gourio F, Noual PA (2009) The marginal worker and the aggregate elasticity of labor supply. Boston University Dept. of Economics Working Papers Series, WP2006-009
Guvenen F (2006) Reconciling conflicting evidence on the elasticity of intertemporal substitution: A macroeconomic perspective. J Monet Econ 53(7):1451–1472
Holmes TJ (1998) The effect of state policies on the location of manufacturing: Evidence from state borders. J Polit Econ 106(4):667–705
Hubbard RG, Skinner J, Zeldes SP (1994) Precautionary saving and social insurance. No. w4884. National Bureau of Economic Research
Imai S, Keane MP (2004) Intertemporal labor supply and human capital accumulation. Int Econ Rev 45(2):601–641
Imrohoroğlu S, Kitao S (2012) Social security reforms: Benefit claiming, labor force participation, and long-run sustainability. Am Economic Journal: Macroeconomics 4(3):96–127
Jaimovich N, Siu HE (2009) The young, the old, and the restless: Demographics and business cycle volatility. Am Econ Rev 99(3):804–826
Judd KL (1987) The welfare cost of factor taxation in a perfect-foresight model. J Polit Econ 95(4):675–709
Kaufman BE (1999) Emotional arousal as a source of bounded rationality. J Econ Behav Organ 38(2):135–144
Kydland F (1984) Heterogeneity in labor market and business cycles. In Carnegie-Rochester Conference Series on Public Policy vol. 21, pp. 173–208
MaCurdy TE (1981) An empirical model of labor supply in a life-cycle setting. J Polit Econ 89(6):1059–1085
Merz M (1999) Heterogeneous job-matches and the cyclical behavior of labor turnover. J Monet Econ 43(1):91–124
Mustre-del-Rio J (2015) Wealth and labor supply heterogeneity. Rev Econ Dyn 18(3):619–634
Peterman WB (2013) Determining the motives for a positive optimal tax on capital. J Economic Dynamics Control 37(1):265–295
Peterman WB (2016) Reconciling micro and macro estimates of the Frisch labor supply elasticity. Econ Inq 54(1):100–120
Pistaferri L (2003) Anticipated and unanticipated wage changes, wage risk, and intertemporal labor supply. J Labor Econ 21(3):729–754
Rogerson R, Wallenius J (2013) Nonconvexities, retirement, and the elasticity of labor supply. Am Econ Rev 103(4):1445–1462

Whalen C, Reichling F (2017) Estimates of the Frisch elasticity of labor supply: A review. East Econ J 43(1):37–42
Zeldes SP (1989) Consumption and liquidity constraints: an empirical investigation. J Polit Econ 97(2):305–346
Ziliak JP, Kniesner TJ (1999) Estimating life cycle labor supply tax effects. J Polit Econ 107(2):326–359

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Authors and Affiliations

Seyedmasood Dastan[1] · Faraz Farhidi[2]

✉ Seyedmasood Dastan
sdastan@utep.edu

Faraz Farhidi
ffarhidi@gsu.edu

[1] University of Texas at El Paso, Department of Economics & Finance, El Paso, TX, USA

[2] Georgia State University, Economics Department, Atlanta, GA, USA